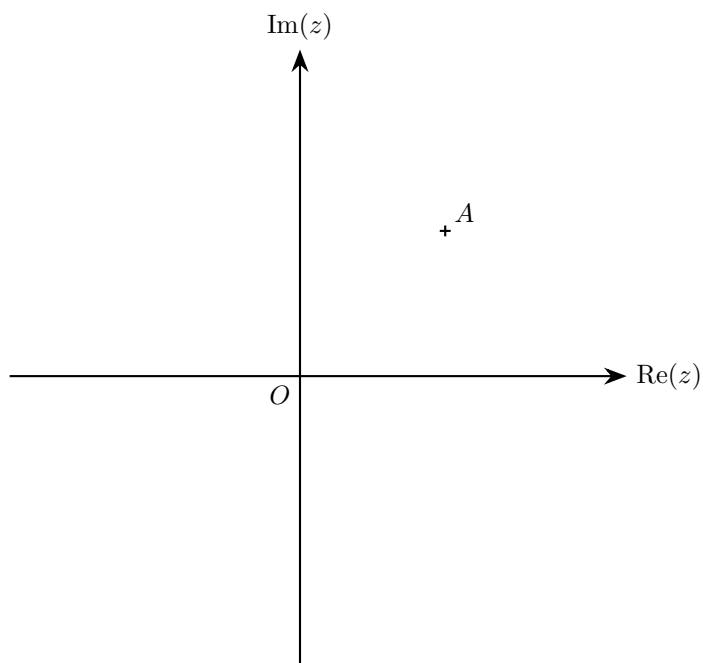


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1. An Argand diagram with the point A representing a complex number z_1 is shown in the diagram. The complex numbers z_2 , z_3 and z_4 are $z_1 e^{\frac{1}{2}\pi i}$, $z_1 e^{\pi i}$ and $z_1 e^{\frac{3}{2}\pi i}$ respectively.
- (a) (i) On a copy of the Argand diagram, mark the points B , C and D representing the complex numbers z_2 , z_3 and z_4 . [2]
- (ii) Show that $z_1 + z_2 + z_3 + z_4 = 0$. [2]
- (b) Given now that z_1 , z_2 , z_3 and z_4 are roots of the equation $z^4 = -16$, find these four roots, giving your answers in the form $a + ib$, where a and b are real and exact. [4]

- 2.** The five solutions of the equation $z^5 = 16 + 16\sqrt{3}i$ are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon P with centre at the origin.

- (a) Determine the five complex numbers that represent these vertices, giving your answers in the form $re^{i\theta}$, where $0 \leq \theta < 2\pi$. [4]
- (b) Sketch P on an Argand Diagram, taking care to label each vertex. [2]
- (c) Show that the area of P can be written $k \sin \frac{2\pi}{5}$ where k is an integer to be determined. [4]

- 3.** (i) Find the 8th roots of $-128 + 128\sqrt{3}i$ in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta < 2\pi$. [5]
- (ii) These 8th roots are represented in the Argand diagram, in order of increasing θ , by the points A, B, C, D, E, F, G, H .
Draw the Argand diagram, making clear which point is which. [2]
- (iii) The mid-point of BD is the point P which represents the complex number w .
Find, in exact form, the modulus and argument of w . [3]
- (iv) w is an n th root of a real number a , where n is a positive integer. State the least possible value of n and find the corresponding value of a . [3]

4. Given that $z_1 = 7 - i$ and $z_2 = \frac{z_1}{1 - i}$,

(a) find z_2 in the form $a + bi$, where a and b are real. [2]

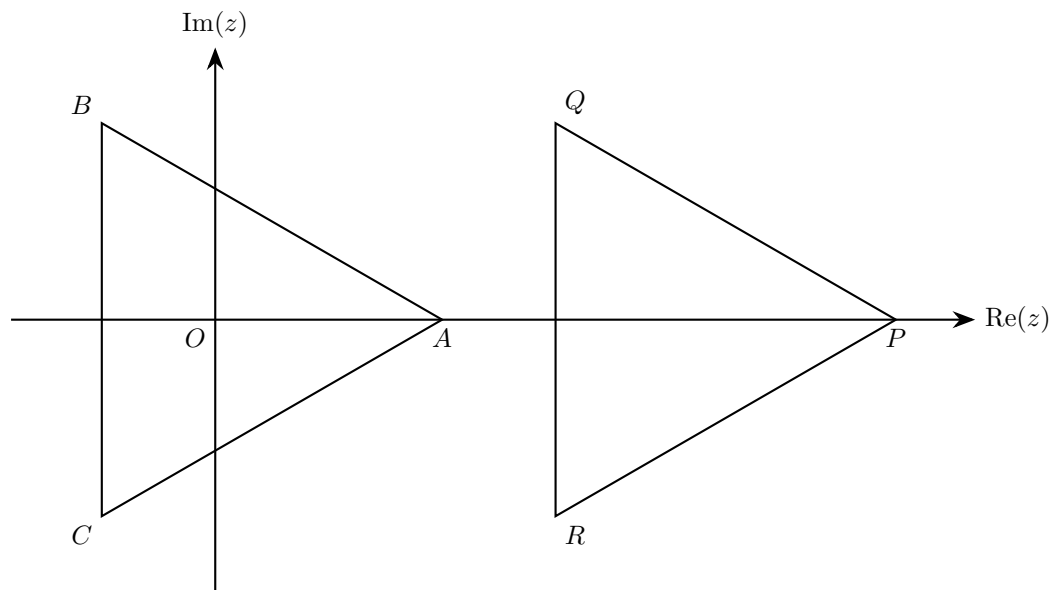
(b) Show on an Argand diagram the point P representing z_1 and the point Q representing z_2 . [2]

(c) Given that O is the origin, show that $\angle OQP = \frac{\pi}{2}$. [2]

The circle passing through the points O , P and Q has centre C . Find

(d) the complex number represented by C . [2]

(e) the exact value of the radius of the circle. [2]



5. The cube roots of unity are represented on the Argand diagram above by the points A , B and C .

The complex numbers represented by P , Q and R are obtained by adding 2 to the complex numbers represented by A , B and C respectively.

Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points A , B , C , P , Q and R .

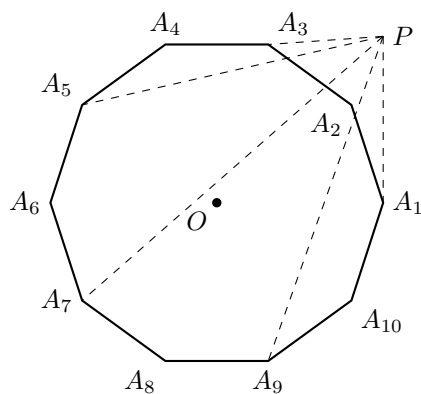
[5]

6. In an Argand diagram, the points A , B and C are the vertices of an equilateral triangle with centre G . The point G represents the complex number $1-2i$, and the point A represents the complex number $5+2i$.

- (a) Find the complex numbers represented by the points B and C , giving your answers in the form $x + iy$, where x and y are real and exact. [6]

The points D , E and F are the centroids of triangles GBC , GCA and GAB respectively.

- (b) Find the exact area of triangle DEF . [4]



7. (a) Using the identity $zz^* = |z|^2$, or otherwise, show that if w is any root of unity then

$$|w - (1 + i)|^2 = 3 - (1 - i)w - (1 + i)w^* \quad [3]$$

The diagram shows a regular decagon $A_1A_2 \dots A_{10}$ whose vertices all lie on the unit circle with centre at the origin O and A_1 at $(1, 0)$. The point P lies in the same plane as the decagon and has coordinates $(1, 1)$.

- (b) Using the result from part (a), find

$$\sum_{r=1}^5 (PA_{2r-1})^2 \quad [4]$$

8. You are given the complex number $\alpha = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$ and the equation $z^5 = -1$.

(a) Show that α is a root of the equation. [2]

(b) Write down the other four roots of the equation. [1]

(c) Show that

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0 \quad [2]$$

(d) Hence show that

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0 \quad [3]$$

(e) Hence determine the value of $\cos \frac{3\pi}{5}$ in the form $a + b\sqrt{c}$, where a , b and c are rational numbers to be found. [4]

9. The points representing the complex numbers $z_1 = 6 + 11i$ and $z_2 = -8 - 3i$ are opposite vertices of a regular octagon, K , in the complex plane.
The centre of K represents the complex number α .

(a) Show that $\alpha = -1 + 4i$. [2]

Given that

$$\beta = \frac{1-i}{14}$$

(b) Show that $\beta(z_1 - \alpha) = 1$ [2]

The vertices of K are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

- (c) (i) Write down the roots of the equation $w^8 = 1$ in the form $re^{i\theta}$. [1]
(ii) Hence, or otherwise, determine the positions of the other six vertices of K , giving your answers as complex numbers in Cartesian form. [4]

- 10.** In an Argand diagram, the points A and B are represented by the complex numbers $1 + 4i$ and $7 + 2i$ respectively. The circle C passes through A and B , and its centre lies on the real axis.

(a) Find the equation of C , giving your answer in the form

$$|z - a| = b \quad a \in \mathbb{C}, b \in \mathbb{R} \quad [4]$$

The circle D , with equation $|z - (1 + i)| = 5$, intersects C at the points representing the complex numbers z_1 and z_2 .

(b) Find the complex numbers z_1 and z_2 . [5]

11. The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where p and q are real, has roots α , β and γ .

It is given that $\alpha = -2 + 2i$.

(a) (i) Write down another root, β , of the equation. [1]

(ii) Find the third root, γ . [3]

(iii) Find the values of p and q . [3]

(b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

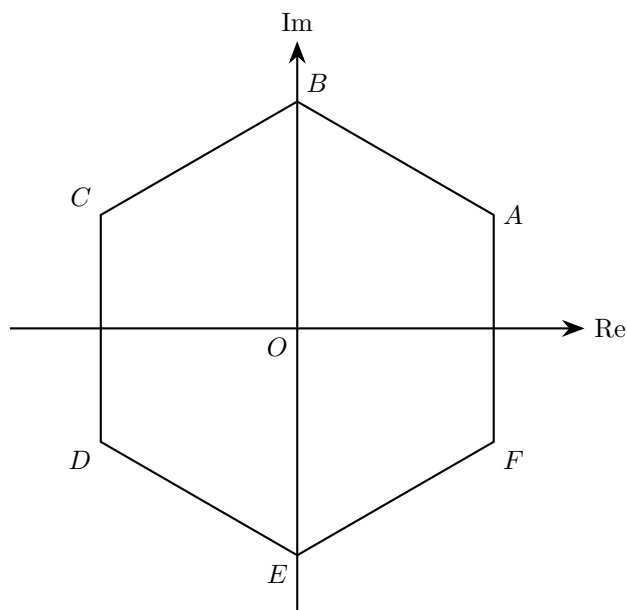
(ii) Hence show that

$$(-2 + 2i)^n = (2\sqrt{2})^n \left(\cos \frac{3n\pi}{4} + i \sin \frac{3n\pi}{4} \right) \quad [2]$$

(iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos \frac{3n\pi}{4} + (-3)^n$$

where n is an integer. [3]



12. In the diagram, the points A , B , C , D , E and F represent the complex sixth roots of -729 on an Argand diagram. The centroids of triangles OAB , OBC , OCD , ODE , OEF and OFA are G , H , I , J , K and L respectively.

- (a) Write down, in exponential form, the complex numbers represented by the points A , B , C , D , E and F . [2]
- (b) When these complex numbers are multiplied by the complex number w , the resulting complex numbers are represented by the points G , H , I , J , K and L . Find w in exponential form. [4]
- (c) You are given that G , H , I , J , K and L represent roots of the equation $z^6 = p$. Find p . [2]

13. (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1 \quad [2]$$

- (ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-i \cot\left(\frac{m\pi}{22}\right) \quad [6]$$

where $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$.

- (iii) Show that

$$\left\{-i \cot\left(\frac{m\pi}{22}\right)\right\} \left\{i \cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right) \quad [1]$$

- (iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right) \tan^2\left(\frac{3\pi}{22}\right) \tan^2\left(\frac{5\pi}{22}\right) \tan^2\left(\frac{7\pi}{22}\right) \tan^2\left(\frac{9\pi}{22}\right) \quad [2]$$